PAUL T. SUMMERS

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EDUCATION

Stanford University, Stanford California — *September 2018 - June 2024*
Ph.D. in Geophysics — GPA: 3.94

Stanford University, Stanford California — *September 2010 - June 2014*
B.S in Physics, M.S. in Geophysics — GPA: 3.81, 3.89

RESEARCH AND PROFESSIONAL EXPERIENCE

Tufts University & Rutgers University & Georgia Institute of Technology — August 2024 - Present
Postdoctoral Researcher — *Atlanta, Georgia*
Numerical modeling of ice mélange and interactions with ocean currents and glacial dynamics.
Implemented and extended existing numerical modeling packages `MITgcm` and `GLACIOME1D` in Fortran, Python on Linux HPC systems.

Stanford University Department of Geophysics — September 2018 - June 2024
Ph.D. Candidate — *Stanford, CA*
Computational fluid dynamics focused on Antarctic Ice Sheet. Analyzed real world data for validating model outputs.
Thermomechanical modeling and ice penetrating radar sounding processing techniques focused on Antarctic shear margins.

Dropbox Inc. — August 2014 - July 2018
Software Developer — *San Francisco, CA*
Designed, built and tested developing software for enterprise scale solutions. Change management tools and unit testing.
Agile software development within regulated financial systems, coordinating across many teams.

Stanford University Department of Geophysics — June 2013 - June 2014
Researcher, M.S. Candidate — *Stanford, CA*
Numerical modeling of fluid mechanics of pre-explosive harmonic tremor in the 2009 Redoubt Volcano eruption.

SELECTED PUBLICATIONS

May, D.F.; Schroeder, D.M.; **Summers, P.T.**; Teisberg, T.O.; Broome, A.L.; Bienert, N.L.; Zak, J.; Young, T.J.; Karplus, M.S.; Tulaczyk, S. "Polarimetric Multi-Offset Radio-Echo Sounding with a Radio Frequency-Over-Fiber ApRES System," *Journal of Glaciology*, 2025, `https://doi.org/10.1017/jog.2025.10114`

Summers, P.T.; Jackson, R. H.; Robel, A. A. "Sub-grid Parameterization of Iceberg Drag in a Coupled Iceberg-Ocean Model", *The Cryosphere*, 2025, `https://doi.org/10.5194/tc-19-5135-2025`

Teisberg, T. O.; Schroeder, D. M.; **Summers, P.T.**; Morlighem, M. "Measurement of Englacial Velocity Fields With Interferometric Radio Echo Sounders," *Journal of Geophysical Research: Earth Surface*, 2025, `https://doi.org/10.1029/2025JF008286`

Summers, P.T.; Schroeder, D. M.; May, D. F.; Suckale, J. "Evidence for and against temperate ice in Antarctic shear margins from radar-depth sounding data," *Geophysical Research Letters*, 2024, `https://doi.org/10.1029/2023GL106893`

Summers, P.T.; Elseworth, C.W.; Dow, C.F.; Suckale, J. "Migration of the Shear Margins at Thwaites Glacier: Dependence on Basal Conditions and Testability Against Field Data," *Journal of Geophysical Research: Earth Surface*, 2023, `https://doi.org/10.1029/2022JF006958`

Bienert, N.; Schroeder D. M.; **Summers, P.T.** "Bistatic Radar Tomography of Shear Margins: Simulated Temperature and Basal Material Inversions," *IEEE Transactions on Geoscience and Remote Sensing*, 2022, `https://doi.org/10.1109/TGRS.2022.3213047`

SELECTED AWARDS AND FUNDING

NSF Office of Polar Programs - Arctic Science — Pending
Lead PI for proposal "Assessing the Stability of Coupled Glacier-Mélange-Ocean Systems Across Greenland" for $337,192

ARCS Scholar — 2022 - 2024
Northern California Chapter of the Achievement Rewards for College Scientists, 2x recipient for total of $101,000

Best Graduate Poster — May 2023
Research Review Symposium — *Stanford Doerr School of Sustainability*
Radar Attenuation Signature of Temperate Antarctic Shear Margins

TECHNICAL STRENGTHS

Computer Languages	Python, MATLAB, Fortran, Bash, Java, SQL, APEX, SOQL, Javascript
Tools	HPC, Git, Gitlab, MITgcm, vim, LaTeX, Sublime IDE, Linux Shell Environments & Scripting Parallel Programming (MPI), HPC Queue Management(SLURM)
Field Skills	ApRES, Seismic Surveying (Ice and Land), ERT, GPS

SELECTED TEACHING AND MENTORING

JIRP Academic Faculty July 2024 - Present
Juneau Icefield Research Program
Led lectures, discussions, mentored student-led projects, and in-field instruction for 32 undergraduate and graduate students.

SESUR Program Assistant April 2022 - October 2022
Stanford Doerr School of Sustainability
Coordinate Stanford Earth Summer Undergraduate Research Program including field trips, weekly seminars, various social events.

OPEN SOURCE CODE REPOSITORIES

Zenodo For Publications
`https://zenodo.org/records/14721713` Model developed for (Summers, et al. 2025)
`https://zenodo.org/records/15116445` Data for (Summers, et al. 2025)
`https://zenodo.org/records/10783426` (Summers, et. al. 2024)
`https://zenodo.org/record/7106136` (Summers, et. al. 2023)

Github
`https://github.com/somonesummers` Ongoing research and personal projects
`https://github.com/somonesummers/demoMITgcm` Idealizde demo for running MITgcm and iceberg2 package

FIELD EXPERIENCE

Juneau Icefield Research Program July 2024 - Present
Academic Faculty *Taku Glacier, Juneau Icefield*
Led week long research projects with 6-8 students. Worked with RTK GPS survey, tiltmeters, passive seismometers. Led field trip based lectures and activities.

Thwaites Interdisciplinary Margin Evolution Oct 2023 - Feb 2024
Field Scientist *Thwaites Glacier, West Antarctica*
Wide offset (up to 4 km) bistatic, polarimetric radar survey using wireless and fiber optic synchronization techniques using modified pRES radar. Assisted with 2-D and 3-D active seismic survey. Surveyed and deployed seismic nodes with RTK GPS, assisted in active seismic explosive sources. 7 weeks in the deep field in a team of 16 with 2 guides.

Near-Surface Geophysics: Imaging Groundwater Systems May 2022
Teaching Assistant *Coyote Valley, California*
Co-led a class of 20 undergraduates to completed a 100 meter seismic (hammer and Betsy gun), 200 m electrical resistivity tomography, and towed transient electromagnetic survey imaging ground water connectivity in the top 40 meters of the subsurface. Worked with community decision makers to inform development of newly acquired public lands.

Thwaites Interdisciplinary Margin Evolution Oct 2021 - Jan 2022
Field Scientist *Thwaites Glacier, West Antarctica*
Completed a 5 km offset bistatic, polarimetric radar survey. Deployed and recovered seismic nodes in an active seismic survey using hammer source. Recovered passive seismic nodes and GPS stations. 3 weeks in the deep field in a team of 4 scientists and 2 guides.

SELECTED CONFERENCE ABSTRACTS

AGU 2025 Dec, 2025
Summers, P. T.; Robel, A. A.; Jackson, R. H. (2025, Dec). Rapid Mélange Collapse and De-Buttressing in a Coupled Glacier/Ocean/Mélange Model (GLACIOME)

AGU 2025 Dec, 2025
Baker, J.; Robel, A. A.; **Summers, P. T.** (2025, Dec). Quantifying the Feedback Between Ice Mélange Buttressing Force and Glacier Calving.

WAIS 2025 Sept, 2025
Summers, P. T.; Robel, A. A.; Jackson, R. H. (2025, Dec). Glacier-Mélange Feedbacks as a Stabilization Mechanism for Classical Marine Ice Sheet Stability.

Kavli Institute for Theoretical Physics: The Future of Earth's Polar Regions Jun, 2025
Summers, P. T.; Jackson, R. H.; Robel, A. A. (2025, Jun). GLACIOME: The Coupled Glacier/Ocean/Melange Model. `https://online.kitp.ucsb.edu/online/polar-c25/`

AGU 2024 Dec, 2024
Summers, P. T.; Schroeder, D. M.; May, D. F.; Suckale, J. (2024, Dec). Constraints on the Thermal State of Antarctic Shear Margins from Integration of Thermodynamic Modeling and Airborne Ice Penetrating Radar Data.

AGU 2024 Dec, 2024
May, D. F.; Schroeder, D. M.; **Summers, P. T.**; Teisberg, T. (2024, Dec). Multi-Offset Fiber Optic-Based Radar Arrays For Time-Lapse Imaging of Englacial and Subglacial Processes.